HEALTHCONNECT RWANDA

Final Analytics & Decision Support Report

Week 8 | Data Analytics Track | Final Report

AnalystLab Africa Experience Lab

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Executive Summary

HealthConnect Clinic is a fictional healthcare provider seeking to reduce missed appointments, improve patient appointment attendance, make better use of available appointment slots and provide more effective administrative support to patients.

The Data Analytics contribution uses the available HealthConnect data to provide management with a view of patient information, appointment activity, clinical records, financial activity, healthcare operations and patient experience. The main output is a six-page Power BI dashboard supported by SQL analysis and a Business Insights Report.

The dashboard reports 20,000 registered patients, 5,000 appointments, 350 doctors, 20 healthcare facilities, total revenue of 1,132,677,500 and average patient satisfaction of 2.99. Key KPI figures were reviewed during Week 7 and compared with SQL results where available.

The analysis identifies areas for management attention, including patient feedback, a declining pattern in the monthly revenue visual, differences between billing, payment and revenue totals, and limitations in operational and clinical reporting. The available appointment CSV contains 5,000 records but has no attendance-status field; therefore, reliable attendance and no-show rates cannot be calculated from that file.

The dashboard provides a useful overview of the available records, but it should be treated as a decision-support tool rather than proof that a particular intervention will automatically improve attendance or patient outcomes.

1. Week 7 to Week 8 Transition

1.1 Main Week 7 output

The main Week 7 output was the HealthConnect Analytics Testing and Refinement work. This involved reviewing the six-page Power BI dashboard, validating key KPIs against SQL query results, testing dashboard filters and interactions, investigating financial figures and refining dashboard visualisations.

1.2 Most important testing results

The appointment count (5,000), doctor count (350), facility count (20), average patient satisfaction (2.99) and total revenue (1,132,677,500) were reported as matching corresponding SQL query results during Week 7. The registered-patient KPI was also reported as validated during the Week 7 review.

1.3 Major issue discovered

The financial investigation identified differences between revenue, billing and payment totals. It also identified billing records marked as Pending that had associated payment amounts. These findings require careful interpretation because the available information does not establish that the measures share the same definition, period or reporting basis.

1.4 Refinements completed

- Reviewed dashboard filters, slicers and visual interactions.
- Reviewed attendance/no-show findings across the segments available in the analysis.
- Improved visual readability, chart labels and blank categories.
- Investigated financial figures and documented remaining questions.

1.5 Week 8 transition status

The existing dashboard and analytical findings are being carried forward for final documentation and presentation. Cross-track testing was not completed during Week 7. No Week 7 cross-track output exchange is claimed in this report.

2. Final Integration Readiness

2.1 Final component and business problem

The final component is the HealthConnect Final Analytics and Decision Support Package: the Power BI dashboard, KPI summary, analytical findings, business insights and recommendations. It addresses the need to understand available patient, appointment, clinical, financial, operational and patient-experience information.

2.2 Current status and outputs

The six-page dashboard has been developed and reviewed. Key KPI figures were checked against SQL results where available; dashboard filters and interactions were tested; and visual presentation was refined. Planned final outputs include the dashboard, final KPI summary, validated findings, key visualisations, business insights, recommendations, analytical limitations, executive summary and presentation materials.

2.3 Dependencies and cross-track status

The Week 8 brief describes relationships between Data Analytics, Data Science and Project Management. For example, validated analytical findings may support model interpretation and project

decisions. However, no actual Week 7 cross-track testing or exchange was completed, and no Week 8 output from another track has yet been confirmed in the materials available for this report. These relationships are therefore described as potential dependencies, not completed integration.

2.4 Readiness evidence

- Six-page Power BI dashboard.
- Week 7 SQL KPI and financial testing results as reported by the intern.
- Week 7 Testing and Refinement Report and KPI/Financial Testing Log.
- Dashboard screenshots and Dashboard Evidence Register.
- Business Insights Report and final presentation materials.

3. Final Analytics and Decision Support

3.1 Final KPI summary

KPI	Final value
Registered patients	20,000
Total appointments	5,000
Total revenue	1,132,677,500
Doctors	350
Healthcare facilities	20
Average patient satisfaction	2.99

The currency associated with the revenue figure is not identified in the available dashboard information.

3.2 Key business insights

Patient experience

Average patient satisfaction is 2.99. The dashboard also presents ratings from 1 to 5 and feedback categories. These can support review of patient experience and identification of areas that may require attention.

Appointment attendance

Attendance and no-show outcomes are central to the HealthConnect business problem. However, the supplied appointment CSV does not contain an attendance-status field, so actual attendance and no-show rates cannot be calculated from that file.

Revenue pattern

The monthly revenue visual shows a declining pattern across the months displayed. The dashboard does not establish the cause; the pattern should be investigated alongside appointment activity, billing and payment information.

Patient characteristics

Age-group, gender, geographical and chronic-condition visuals can help management understand the profile of registered patients and consider service planning.

Operational reporting

The operations dashboard provides recorded staff, bed and equipment information. Equipment counts represent equipment records, not usage. The bed visual should not be interpreted as complete hospital-wide availability without further verification.

Financial definitions

Revenue, billing and payments totals differ and should not be treated as interchangeable without confirming their definitions, reporting periods and calculation methods. Pending billing records with associated payment amounts also require investigation.

3.3 Business recommendations

- Introduce a structured appointment outcome field with values such as Attended, No-show, Cancelled and Rescheduled, and use it to monitor attendance consistently.
- Establish a structured appointment follow-up process to help reduce missed appointments once reliable attendance data is available.
- Review patient feedback regularly and investigate facilities or departments where ratings are lower.
- Investigate the declining monthly revenue pattern before drawing conclusions about its cause.
- Clearly distinguish billed amounts, payments received and revenue in financial reporting.
- Improve facility and department reporting through descriptive names and consistent data definitions.
- Expand clinical reporting by capturing transaction-level laboratory tests, prescriptions, admissions and referrals.
- Record equipment status and utilisation and maintain complete bed-status information.

4. Analytical Limitations and Risks

Appointment data

The supplied appointment CSV contains 5,000 records and fields for AppointmentID, PatientID, DoctorID, FacilityID, DepartmentID, Date and VisitType. It does not contain an attendance-status field. Reliable attended, no-show, attendance-rate and no-show-rate measures therefore cannot be calculated from this file.

Financial data

Revenue, billing and payment totals differ. Their definitions and reporting periods have not been established as directly comparable. Pending billing records with associated payment amounts require further investigation.

Facility identification

Some facility comparisons use facility IDs because descriptive facility names were unavailable in the inspected data.

Clinical detail

Separate transaction-level records for laboratory tests, prescriptions, admissions and referrals were not available in the inspected tables.

Operations

The equipment information represents recorded equipment rather than actual utilisation. Bed-status information requires verification before it can be interpreted as overall hospital availability.

Patient feedback

The feedback table had no visible date field, limiting analysis of feedback or complaint trends over time.

Cross-track collaboration

Week 7 cross-track collaboration was not completed.

5. Conclusion

The Data Analytics contribution brings together the available HealthConnect healthcare information through a six-page Power BI dashboard and supporting analysis. Week 7 testing and refinement helped review key indicators, improve presentation and identify data limitations. The dashboard can support

management review of patient, appointment, clinical, financial, operational and patient-experience information.

A key priority for future improvement is to capture appointment outcomes so attendance and missed appointments can be measured reliably. Clearer financial definitions, richer clinical records and improved operational data would also strengthen the decision-support solution. The findings should be interpreted within the limits of the supplied data.